

State of the art review on model predictive control (MPC) in Heating Ventilation and Air-conditioning (HVAC) field

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ABSTRACT

Building systems are subject of dynamic system that have a general feature of non-linearity and in turn, present us with different challenges for its optimized control of energy-saving and thermal comfort. Occupancy behavior, weather forecast, ambient temperature and solar irradiation, etc. In particular are difficult to predict. These uncertainty parameters have a direct influence on the building's behavior that further complicate problem formulation for energy saving in a building. Model predictive control (MPC) has been one of the potential strategies for control schemes to address these problems and tackle them since its invention. MPC is a suitable and the best candidate when it comes to questioning for future predictions in terms of energy efficiency, cost, and control mechanisms. MPC consists of model of a plant, prediction horizon and optimization tools used for the optimization of the future response of the plant. After broad applications of MPC in industrial applications for process control, it has been gaining ground in the field of Heating Ventilation and Air-conditioning (HVAC). Although there has been extensive research of MPC in HVAC systems of buildings, there lacks a detailed review, a complete structure that formulates and describes the applications. An overall holistic view of applications of MPC in building HVAC system has been provided in this paper. Broader information on modeling techniques and optimization algorithms are discussed in a detailed manner. Different design parameters such as prediction horizon, time step, cost function, etc., that ultimately affect MPC performance are presented in a comprehensive form. Various kinds of modeling software with their technical features, pros and cons are elaborated. The main goal of the current paper is to highlight important design parameters crucial for the MPC control scheme and provide better guidelines for further studies. Various future outlines have been listed that can be helpful for future work in this field.

1. Introduction

A significant amount of energy consumptions of a building comes from the cooling and heating system of buildings and is promising candidates for carbon emission. Energy management in building HVAC systems will be a crucial factor in the search for solutions related to global energy demand. A lot of hard staking efforts have been put forward by academicians and control industries in designing control approaches for efficient energy management in buildings. Some of the major issues primarily faced during the implementation of the control technique in the HVAC system are summarized as below:

- Dynamical behavior of the system.
- Uncertainty in disturbance predictions.
- Practical feasibility and applicability.

- Integration of renewable energy system (RES) such as smart-grid and microgrid with the HVAC system.
- Non-linearity and complex nature of HVAC system.

Classical approaches (typically like PIDs) of HVAC control are the most sought out technique due to their practical feasibility. These techniques, however, focus only on indoor environment conditioning rather than efficient control approaches. In contrast to the classical control, model predictive control (MPC) is a promising candidate for smart buildings, efficient control, higher energy savings, better indoor environment, etc [1]. MPC is an advanced form of a control strategy for process control which has underlying principles for satisfying the constraints [2]. It consists of a plant model, optimizer solver, and a prediction horizon for generating control trajectory for a process. MPC has been introduced late in process control in the early 80s. One of the main reasons for its late introduction in the real-life system is due to a lack of

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Nomenclature

AHU	Air handling unit
ANN	Artificial Neural Network
ARX	Autoregressive exogenous inputs
ADMM	Alternating Direction Method of Multipliers
AMPC	Adaptive model predictive control
ARMAX	Autoregressive–moving-average model with exogenous inputs
ASHRAE	American Society of Heating, Refrigerating and Air-conditioning Engineers
BAS	Building Automation System
BIC	Bayesian information criterion
BMS	Building management system
CO ₂	Carbon dioxide
CL	Closed loop
CAV	Constant air volume
CL-MPC	Centralized model predictive control
DMPC	Distributed model predictive control
EMS	Energy management system
FDD	Fault detection and diagnostics
FTC	Fault tolerant control
FDI	Fault detection and isolation

GA	Genetic algorithm
GHG	Green house gas
HMPC	Hybrid model predictive control
MOO	Multi-objective optimization
MIMO	Multiple input multiple output
MOGA	Multi-objective genetic algorithm
NN	Neural Network
QP	Quadratic programming
PSO	Particle swarm optimization
PMV	Predicted Mean Vote
PPD	Predicted percentage of dissatisfied
PI	Proportional integral
PID	Proportional integral derivative
R–C	Resistance- Capacitance
RBC	Rule based controller
RMPC	Robust model predictive controller
RNN	Recurrent neural network
SLP	Sequential Linear programming
SQP	Sequential quadratic programming
SEB	Smart energy buildings
SMPC	Stochastic model predictive control
TOU	Time of Utility
VAV	Variable air volume

high advanced processing and computer software.

MPC has various technical specialties and features making it one of the prominent choices for control strategies in building systems. Over 2000 industrial installations with MPC have been the most widely implemented technique in process control strategies [3]. Effectiveness of MPC increases while considering thermal energy storage such as passive building thermal mass or active systems such as ice and chilled water storage [4]. MPC not only saves energy but also improves control performances including control precision, stability, and immunity to interference. They are generally employed where the requirements of control action are very strict. The statistics of important studies on MPC in the last decade is shown in Fig. 1 which means the increasing interests in MPC study by people.

Afram et al. [5] have commenced the studying and reviewing of

MPCs in the HVACs control system with their detailed explanation both in theory and applications. In their study, different parameters affecting the performance of the MPC controller with the best illustrative example have been discussed, and different control strategies applied to build energy management systems have been investigated. Hilliard et al. [6] have also reviewed one specific topic of MPCs for commercial buildings providing broader information on better opportunities and potentials. This article reviewed 19 case studies and compared it with previous control strategies considering different design parameters such as optimization algorithm, time step, sensor information to inputs, etc. Mirakhorli et al. [7] reviewed occupancy-based MPC for controlling the indoor environment. The authors tried to address the stochastic nature of occupancy rate which has a crucial role in the indoor climate through discussing a few experimental studies. Shaikh et al. [8] gave an idea

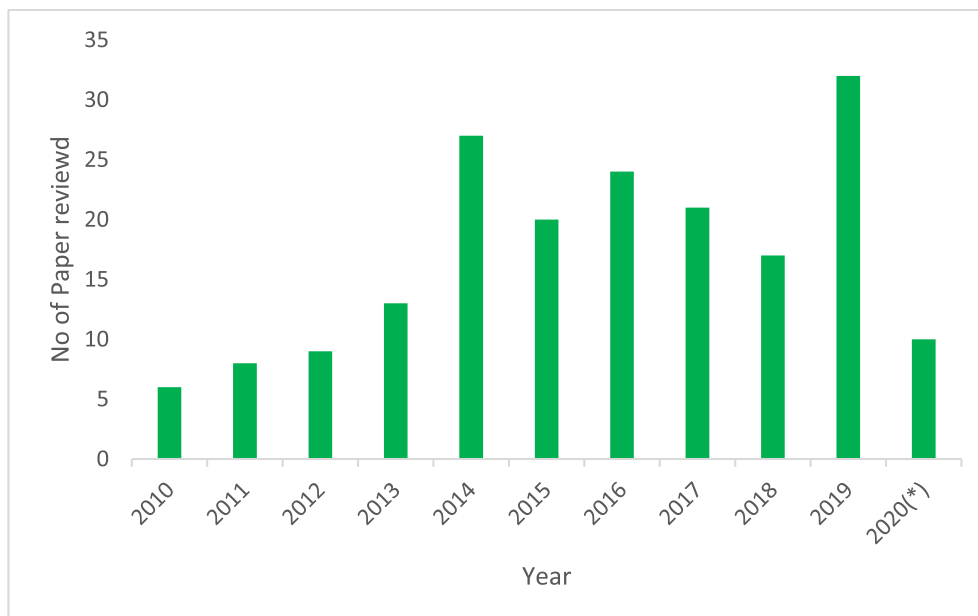


Fig. 1. Numbers of paper reviewed in this literature review.

about different types of the intelligent control scheme for energy and comfort management in smart energy buildings (SEB). Artificial neural network (ANN) based MPC was reviewed and a case study was also formulated using the ANN modeling technique by Afram et al. [9].

Modeling is the key step for MPC implementation, and modeling techniques used for plant design of building HVAC control systems were reviewed in Ref. [10]. In this paper, applicability, ease of acceptance in practice summarizes the strengths, weaknesses, applications, and performances of these modeling techniques. Different types of modeling ideas being used for HVAC systems and a broader classification of the data-driven model have been presented in Ref. [9]. Harish et al. [11] summarized significant methodologies for modeling and simulations of building energy systems and various kinds of building simulation software employed in this field. Killian et al. [12] addressed several types of questions concerning the implementation and problems while designing an MPC control scheme for energy-efficient buildings. In a recent paper [13], the existing MPC algorithms for building and HVAC systems have been reviewed and discussed.

Although a good number of review articles have been published on the topic of MPC, they didn't present a broader picture on variants of MPC. Exhaustive studies of variants and their applicability under different criteria are yet to be conducted. Survey of modeling techniques with their pros and cons, classification of optimization methods, constraint, design parameters of MPC is not addressed yet in an extensive manner. These review studies lack of extended discussions on different simulation platforms and their integration in MPC for smooth operations. This paper distinguishes itself from the previous publications by addressing these issues and provides a broader picture of different aspects of MPC operations. Moreover, these gaps are to be highlighted by synthesizing more recent published articles and providing crystal clear picture on different aspects related to this control scheme.

An extensive review work was conducted with the help of search engines like Google scholars, Scopus, IEEE digital library and some journals of Elsevier, energies, etc. This exploring method was categorized into I) Peer-reviewed Journals and technical papers related with the application of MPC for building HVAC system. II) Books and thesis dissertation. The keywords used for collecting the databases of above materials were "Model predictive control in HVAC", "Model predictive control in VAV System", "Control methods in HVACs", "MPCs vs PID", "building simulation softwares for HVAC" etc. These keywords were explored on the search engines stated above through iterative process of study. These reference articles were studied and analysed properly for proper formulation and layout of this review article. In general, the brainstorming in review process has been emphasized for MPC problem formulation, modeling techniques, optimization algorithms, and simulation softwares.

The main objective of this article is to provide a clear layout of the implementation of the MPC scheme in building HVAC systems. A detailed discussion of MPC variants and their applicability under different situations are presented extensively. Broader insight on the design of elements of MPC for smooth and efficient control action are discussed in an exhaustive manner with enough past literature. This article will also help in pointing out the current research progress and shortcomings related to MPC yet to be identified.

The layout of this paper is as follows: Section 2 deals with theoretical and mathematical formulations for the MPC scheme. This chapter further classifies different variants of the MPC scheme with its proper theoretical and practical illustration. Section 3 discusses key elements of MPC including optimizing goals, modeling, constraints, and design parameters that are required to capture the dynamics of the HVAC system. Section 4 provides a framework for the implementation of the MPC scheme particularly in building the HVAC system. At last, Section 5 deals with major findings from published articles that are discussed in a detailed manner. This part will also help future research scholars as guidelines for those having a keen interest in this field and control

scheme.

2. Types of MPC

Model predictive control (MPC) is rather a new control method that emerged during the late 80s in the process control of chemical industries. MPC strategies can be implemented using constraints for multiple-input and multiple-output (MIMO) systems and have been applied to building energy controls with objective goals like energy minimization, time of utility (TOU), etc. Li et al. [14] developed an Elman-neural network model with a prediction control scheme to solve the problem. Kuboth [15] et al. investigated experimental adaptive MPC with realistic weather data inside single-family house for minimizing the operational costs of heat pump. Pantazaras et al. [16] controlled and predicted future CO₂ concentrations by including it as one of the design factors in the prediction model. This incorporation may result in the exploration of different optimal schemes in building with uncertain behavior toward occupancy rate.

MPC schemes are featured by.

- Multi-variables can be easily dealt such as temperature set point, flow rate set point, valve position etc. [17–21].
- It introduces feed forward control to compensate the measurable disturbances factors such as climatic predictions, occupancy rate, etc. [18,19].
- Minimization of objective cost function can be calculated within well-defined constraints [22–25].
- Easy-to implement control-law i.e. defined optimization problem and different type of solving techniques such as quadratic programming (QP), linear Programming (LP), dynamic programming (DP), genetic algorithm (GA), particle swarm optimization (PSO), etc. can be used to solve defined problems [26–34].

One major concern for the MPC control scheme is a discrepancy in the model and other drawbacks include more computation time in comparison to the other control methods.

The MPC control schemes are capable of minimizing cost and addressing the problems of uncertainty due to disturbances factors within the defined range, which can be expressed normally by Eq. (1) through (5) as below:

$$\min \sum_{k=0}^{N_p-1} a(x(k), u(k), r(k)) \quad \text{Objective Function} \quad (1)$$

Subjected to,

$$x_o = x \quad \text{Initial State} \quad (1a)$$

$$x(k+1) = f(k)x(k) + g(k)u(k) + h(k)r(k) \quad \text{Update State Variables} \quad (1b)$$

$$a(k) = O(x(k), u(k), r(k), c(k)) \quad \text{Output Variables} \quad (1c)$$

$$(x(k), u(k), r(k)) \leq 0 \quad \text{Constraints} \quad (1d)$$

where, k is sampling time, N_p is prediction horizon, $x(k)$ is system state, $u(k)$ is control input, $r(k)$ is vector of disturbance factor of the system $a(k)$ is system output on the system. State vectors, control inputs and disturbances are subjected to constraints. At each time step, MPC scheme compute control by solving an open-loop optimization problem for the defined prediction and control horizon as shown in Fig. 2.

There have been several kinds of MPC including robust model predictive control (RMPC), stochastic model predictive control (SMPC), distributed model predictive control (DMPC), adaptive model predictive control (AMPC) and hybrid model predictive control (HMPC). These variants of MPC are discussed and reviewed in this section.

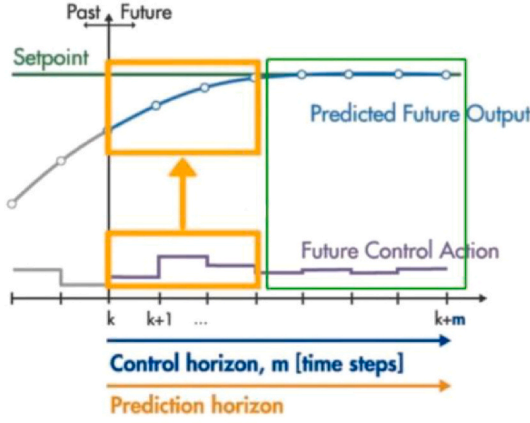


Fig. 2. MPC horizon.

2.1. Robust model predictive control (RMPC)

MPC scheme uses the model of the plant to anticipate future states of the system. The scheme is also feedback control and can perform online optimizations. These features in turn make it a suitable candidate for problems involving performance metrics and hard constraints. However, model uncertainty and coupled subsystems raise serious concerns to this scheme: robustness and scalability. A dual combination of uncertainty and constraint makes robustness a serious concern. In building and HVAC system, accurate modeling cannot be obtained either due to model discrepancy or disturbance factors. Disturbance factors are of exogenous nature that arise generally from cooling loads, outdoor temperature, solar irradiance, and occupancy rate, etc. Thus, uncertainties are troublesome factors for the MPC control scheme resulting in poor accuracy and proper implementation in real systems. The problems may be solved by designing robust model-based predictive control (RMPC) scheme. RMPC is an improvised version of nominal MPC that ensures the state control constraints are satisfied for all possible uncertainty sequences. Here, uncertainties are assumed to be bounded in constraint problem for RMPC formulation. The mathematical formulation of RMPC is expressed as below:

Linear system with system dynamics

$$x_{k+1} = Ax_k + Bu_k + w_k \quad (2)$$

where x is the state vector, u is the input, w is the disturbance vector.

The disturbance lies in a bounded set but is otherwise unknown

$$\forall k \ w(k) \in W \in \mathbb{R}^{N_x} \quad (3)$$

$$y_k = Cx_k + Du_k \quad (4)$$

$$y_k \in \mathcal{Y} \in \mathbb{R}^{N_y} \forall k \quad (5)$$

For output system with constraints, the cost function can be written as:

$$\min \sum_{k=0}^{\infty} p(x_k, u_k) \quad (6)$$

where, y is the output vector; A is the state matrix; B is the state matrix; C is the output matrix; D is the feed-forward matrix; $p(\cdot)$ is stage cost function expresses either in form of linear quadratic or infinity norm.

Some researchers have explored the robust model-based predictive control (RMPC) scheme and its applications. Nagpal et al. [35] adopted RMPC for building climate control in the presence of parametric uncertainties. This control algorithm was successful in keeping the indoor temperature of building in the desired range even in the presence of large disturbances factors. Numerical and simulation studies were made

to validate the results for this control scheme. Yang et al. [36] implemented adaptive robust model predictive control (ARMPC) for refinement of building models and handling the uncertainty of disturbances. This adoption resulted in the variation of energy savings approximately 15%–20% in comparison to thermostat despite the increment in disturbances from 0% to 60%. On the other hand, MPC without adaptive modeling and robust optimization achieve only about 20% energy savings as uncertainty level increases but at the expense of compromising thermal comfort. Therefore, these results further validate that modeling and robust optimization lead to the prevention of indoor conditions from violating constraints due to model inaccuracy and uncertainties in measured disturbances. Huang et al. [37] applied the RMPC scheme to variable air volume (VAV) for zonal control and further demonstrated that the RMPC scheme was able to enhance robustness without much user intervention and could effectively deal with uncertainty factors like cooling load, disturbance factors. Setayesh et al. [38] estimated indoor temperature and humidity with a hybrid control scheme using full and minimum order observers. The authors investigated the effect of model uncertainty factor on air handling unit (AHU) performance and control inputs. Xinhua et al. [39] employed the RMPC scheme for temperature control of an air-conditioning system. The results were compared with PID and manifested that the proposed strategy was able to deal with constraints and allowed stable and robust control while maintaining acceptable thermal comfort. Zhao et al. [40] designed an off-line RMPC scheme for efficient temperature control of the VAV air-conditioning system. The authors observed that the RMPC scheme could enhance the robustness and satisfy the constraints through simulation results. In summary, a key issue while designing and implementing a robust MPC algorithm is to achieve the desired robustness guarantee without either being too cautious or making the computation significantly more complex.

2.2. Stochastic model predictive control (SMPC)

Robust control approaches consider bound on disturbances neglecting model uncertainty, which means it pessimistically considers the worst case. This worst-case scenario rarely occurs in real situations and may degrade the result of the control scheme badly due to the need for protection against low probability uncertainty. Thus, Stochastic model predictive control (SMPC) is regarded as an alternative of RMPC that considers the probabilistic nature of uncertainties and provides additional information about the uncertainty. SMPC is formulated with chance constraints that eliminate the worst-case scenario of the system and takes into consideration the most likely disturbances that may be encountered in practice. In comparison to the RMPC scheme, the SMPC scheme can even deal with unbounded disturbances and is the most suitable control algorithm for HVAC systems as climatic information and cooling loads and HVAC systems cannot be predicted accurately. The formulation of SMPC problem is expressed by Eq. (7).

$$\min \left[E_w \sum_{k=0}^{N-1} l(y(k) - r(t+k), u(k)) \right] \quad (7)$$

Subject to

$$x(k+1) = f(x(k), y(k), w(k)) \quad (7a)$$

$$y(k+1) = g(x(k), y(k), w(k)) \quad (7b)$$

$$u_{\min} \leq u(t+k) \leq u_{\max} \quad (7c)$$

$$y_{\min} \leq y(k) \leq y_{\max} \quad (7d)$$

$$w(t) \in \{w_1, w_2, w_3, \dots, w_s\}, p_j(t) = \Pr[w(t) = w_j], \sum_{j=1}^s p_j(t) = 1, \forall t \geq 0 \quad (7e)$$

$$x(0) = x(t) \quad (7f)$$

where, the variable representing $x(t)$ is the process state, $y(t)$ is the controlled output, $u(t)$ is manipulated variables, $w(t)$ is disturbance inputs and $\Pr$ denotes probability. In particular, disturbances are often low-dimensional and include electricity price, noise, weather etc., and chance constraints include bound on thermal comfort, temperature setpoints etc.

Different underlying ideas and broader classifications of available techniques on the SMPC were reviewed in terms of performance index, dynamic characteristics, probabilistic constraints [41]. Ma et al. [42] presented a non-linear stochastic model predictive control approach for building temperature regulation. This article optimized energy consumption while bounding to thermal comfort violations using the prediction of weather and occupancy. Separately, Ma et al. [43] presented an SMPC design that has bounds on the probability of thermal comfort violations to minimize energy cost. They estimated and modeled uncertain load of each thermal zone by finitely supported probability density functions (pdfs). Furthermore, Mady et al. [44] simulated the SMPC scheme for minimizing energy consumption and thermal discomfort based on occupancy predictions as control inputs. The result proved that SMPC can achieve a reduction of energy usage by 4.3% while improvising user comfort by an average of 38.3% in comparison to schedule MPC. Long et al. [45] formulated SMPC based on a model of the thermal process, weather condition, occupancy, etc. to regulate the temperature of rooms with economic objectives. Yu et al. [46] formulated a mixed-integer multi time scale SMPC problem for scheduling loads of different characteristics. This study manifested that an energy saving of 30% could be achieved with an SMPC scheme compared with the baseline controllers. In another literature [47], constrained comfort level was maintained at the cost of minimum energy consumptions through the SMPC scheme in which disturbances were predicted through Copulas rather than making assumptions on the distribution of disturbances. In summary, the SMPC schemes are somewhat complicated and solved basically by approximating or defining a specific set of rules, and this also increases the burden on the computational process.

2.3. Distributed model predictive control (DMPC)

MPC scheme is implemented in a centralized network form, and calculations of states are optimized online at once. However, this approach is tedious and cumbersome for large scale applications such as multizone buildings, water distribution systems, etc. Thus, a new control scheme namely a distributed model control scheme (DMPC) has been put forward. The DMPC scheme can split the large-scale applications into smaller and simpler sets of subproblem regulated locally by their input and output is demanded. The motivation behind the DMPC is to reduce computational burden through proper implementation of controllers that are effective to carry out calculations parallelly and communicate efficiently to achieve the objective of feedback systems. The DMPC is the most used control algorithm for building complex consisting of multi-level zones, rooms, and the floor that may ultimately lead to an increase in computational effort. In general, the DMPC is the most suitable candidate for decoupling different equation that is laid out from thermodynamically processes in multi-zone buildings. The mathematical formulation of DMPC is as follows.

$$\min \sum_{i \in M} \left(\hat{V}_i + \sum_{t \in T} l_i(x_{N_{i,t}}, u_{i,t}) \right) \quad (8)$$

subject to

$$x(i, t+1) = A(N_i)x(N_{i,t}) + B(i)u_{i,t} \quad (8a)$$

$$(x(N_{i,t}), u_{i,t}) \in X_{N_i}^* u_i \quad (8b)$$

$$x_{i,t} \in \hat{X}_{f,i} \quad (8c)$$

where, $() = ; f = f_1 * \dots * f_n$; M ; f_i ; $i = 1 \dots n$; α_i $i \in M$; x_i denotes states of the i th zone; u_i denotes control inputs; M denotes dynamically coupled subsystems; $\hat{V}_i$ denotes value function; α denotes positive definite matrix; α denotes decision matrices and f denotes terminal set for decoupled structures.

Some researchers managed to apply the DMPC to the field of HVAC. For example, Moroşan et al. [48] regulated the temperature of the single-zone building with the help of a DMPC control scheme. They also extended the same scheme for multi-zone buildings and validated results with simulations. Results proved that drastic improvement in energy consumption could be achieved through transitions between occupied and unoccupied phase. Scherer et al. [49] presented a distributed model predictive (DMPC) scheme for the efficient management of energy distribution in buildings. A novel strategy was presented where standard counterparts needed strong couplings among the distributed controllers, invariably incurring excessive communication between control agents for effective management of energy distributions in buildings. Thus, DMPC deployed a network of distributed controllers with reduced couplings and less stringent communication needs. Walker et al. [50] developed the DMPC scheme for CO₂ concentration and temperature regulations in three-zone buildings. Their study indicated that less computational effort and nearly the same accuracy achieved using DMPC. Kane et al. [51] developed a DMPC scheme with objectives of maintaining thermal loads at safe temperature and minimization of pump power. The results showed that parallel computations improved computational scalability and added resilience to component failure. Putta et al. [52] conducted a simulation-based study of the DMPC scheme on a medium-sized commercial building. This study proved that a significant reduction of computational time can be achieved in comparison to CMPC while maintaining thermal comfort. Long et al. [53] proposed two-level controllers, i.e. the upper-level controller controls temperature and predicted information to generate trajectories, and the lower level controller used local information to track the reference and optimized energy efficiency and thermal comfort. They implemented the DMPC for maintaining the temperature of each room and zone with minimum energy consumptions. Eini et al. [54] designed a DMPC strategy for maintaining temperatures of each room with the minimum costs of the HVAC system. The results proved that the DMPC strategy achieved 25.42% less energy consumption, 48% less computation time with less tracking error in comparison to centralized MPC(CMPC). In summary, the DMPC schemes provide the better deal between the potential improvement in controller performance and the prediction errors.

2.4. Adaptive model predictive control (AMPC)

Traditional MPC performance and accuracy get reduced drastically while modeling non-linear plants with LTI as thermodynamically processes in building HVAC systems are non-linear and transient. The adaptive model predictive control (AMPC) scheme may be a suitable candidate for handling such problems. The AMPC is better known for handling model uncertainties and updating models based on data measurements extracted online. This scheme has two primary goals: (i) stability of the system must be ensured and (ii) update time-variant models at each time intervals.

An AMPC can be formulated as below:

$$\min \sum_{i \in M} \left(\hat{V}_i + \sum_{t \in T} l_i(x_{N_{i,t}}, u_{i,t}) \right) \quad (9)$$

subject to

$$x(i, t+1) = A(N_i)x(N_{i,t}) + B(i)u_{i,t} \quad (9a)$$

$$(x(N_{i,t}), u_{i,t}) \in X_{N_i}^* u_i \quad (9b)$$

$$x_{i,t} \in \hat{X}_{f,i} \quad (9c)$$

where, $() = ; f = f_1 * \dots * f_M; f_i = i^{n_i} \alpha_i; i \in M; x_i$ denotes states of the i th zone; u_i denotes control inputs; M denotes dynamically coupled subsystems; $\hat{X}_{f,i}$ denotes value function; $\hat{X}_{f,i}$ denotes positive definite matrix; α denotes decision matrices and f denotes terminal set for decoupled structures.

Only a small number of studies have been made on the AMPC scheme in the field of HVAC. For example, Tesfay et al. [55] presented an AMPC mechanism to address these problems where the parameters are tuned continuously through recursive estimation and update approaches making the MPC insensitive to prediction errors and to achieve the optimal superheat response. The authors implemented an adaptive set-point hunting algorithm so that the system could achieve stability and improves energy efficiency simultaneously. Lauro et al. [56] presented an innovative MPC control strategy for indoor temperature regulations in three-zone buildings. The authors presented an idea of ensuring temperature regulation based on the occupancy level of each building zone using two different strategies of AMPC for achieving higher performances. The results indicated the AMPC was superior to the other non-AMPC strategies in terms of computational costs and control performance. Lindelöf et al. [57] applied the AMPC control algorithm to a heating system aiming at maintaining thermal comfort and energy savings. The study results showed that heating energy savings of about 24%–32% were achieved by using the AMPC control for different test sites. En OOI et al. [58] integrated AMPC into the experimental smart house for temperature reference tracking. Their study verified that the output errors of temperature tracking of AMPC were lower than nominal MPC at the cost of increment in computational value. Tanaskovic et al. [59] simulated AMPC for handling model uncertainties of building parameters. The results showed that the proposed AMPC scheme was more efficient in handling uncertainty than nominal MPC and achieved better thermal comfort during the adaption phase.

2.5. Hybrid model predictive control (HMPC)

The recent complexity of model dynamics, goals, and constraints in the problem formulation of MPC has increased rapidly. This system dynamics might fluctuate over the control and prediction horizon, and the variables involved in the optimization problems may be in integral or binary form (i.e. continuous or discrete form). In general, the term hybrid is used particularly when the plant model exhibits certain discontinuity or jump over horizons. These features of MPC formulation fall under the domain of hybrid model predictive control. In general, Hybrid model predictive control (HMPC) is a novel intervention paradigm of the MPC scheme controlled through continuous-valued and discrete-valued state components. This class of MPC helps in its capabilities expansion for dynamics switching, control variables (i.e. mixed integer or binary integers) and logical states. For instance, in the HVAC control system, continuous-valued represents scheduling problems like temperature and airflow and discrete-valued state components would represent logical mode such as on/off, min/max, etc. These types of control schemes are particularly suited when plant models, disturbances factors are uncertain and suffer from poor accuracy. These control variables led to the non-linear optimization problem that is solved using optimization algorithm technique such as mixed integer linear programming (MILP) and mixed integer quadratic programming (MIQP). In general, hybrid MPC are categorized into two special classes i.e. pure discrete system and combination of continuous and discrete system.

A HMPC model can be formulated as follows:

$$\text{min}_{x,i,w} \quad (10)$$

subject to

$$x(0) = x_0 \quad (10a)$$

$$x(N) \in X_f \quad (10b)$$

$$x_{k+1} = A_{i(k)}x_k + B_{i(k)}w_k + f_{i(k)} \forall k \in N_{<n} \quad (10c)$$

$$y_k = C_{i(k)}x_k + D_{i(k)}w_k \forall k \in N_{\leq n} \quad (10d)$$

$$x_k \in \Omega_{i(k)}, w_k \in W_{i(k)}(x_k), i_k \in S, \forall k \in N_{\leq n} \quad (10e)$$

where, $g(\cdot)$ denotes cost function; x denotes state vector; w denotes control input; S denotes set of positive matrices A, B, C, D ; Ω is set of polyhedra; X_f denotes terminal constraint set; y denotes output vector; k denotes time step; N denotes set of natural number; and n denotes prediction horizon.

The HMPC has also been applied to the field of HVAC. Fiorentini et al. [60] conducted practical implementation, designing, and experimentation of the HMPC algorithm on solar-assisted HVAC systems for thermal energy management of the house. The authors achieved high performance using the HMPC method to optimize complex multimode HVAC systems in residential buildings, illustrating efficient selection of the appropriate operating modes to optimally manage thermal energy of the house. Mayer et al. [61] presented branch and bound approach for HMPC of a building cooling supply. The proposed strategy showed excellent performance when compared with implemented conventional control schemes resulting in increased usage of at least 50% of renewable energy sources, a cut in energy costs of about 50%, and an optimized operation of the active storage while accurately delivering the prescribed cooling power trajectory. Huang et al. [62] designed HMPC scheme to minimize energy expenses and the running cost of HVAC systems in an airport building. The results showed that performed HMPC achieved energy savings of about 41% and cost savings of about 13% as compared to baseline control. In Ref. [63], HMPC was proposed and simulated for bilinear dynamic model of stratified thermal storage tanks in buildings to predict cold demand with minimum costs. The results showed that this algorithm was practically feasible for buildings with a limited number of heat/cold generators as it required fewer data and lower costs for instrumentation. Sangi et al. [64] implemented three different types of control mode i.e. HMPC, agent-based controller, and ON/OFF controller to save energy consumptions by applying the concept of multi-agent systems. The result showed that energy consumption for the HMPC scheme was 2.06% and 1.31% lower than the ON/OFF and the agent-based controls, respectively. Khakimova et al. [65] implemented the HMPC scheme in small-sized buildings to regulate room temperatures at minimal electricity costs. The results showed that the proposed control schemes minimized the expense for the consumer without violating large constraints.

In summary, HMPC increases the stability of the control algorithm by avoiding harsh actions. Their problems are mixed-integer problems, thus mostly optimized through DP, MILP, and MIQP algorithm.

3. Performance parameters of MPC

MPC scheme consists of (a) prediction model (b) objective function for establishing optimization problem. (c) defining control law with constraints. A variety of options can be chosen for each element giving rise to different algorithms.

3.1. Modeling

Model predictive control as its name implies requires appropriate modeling of systems and components for controlling the dynamic processes in the building and HVAC system. The model is the cornerstone of the MPC system, which is used to predict the future process dynamics of the controlled plant. The schematic diagram of MPC integration in

building HVAC systems is shown in Fig. 3. Inputs for building the HVAC system are thermal comfort, utility price, occupancy rate, and climatic information. Modeling of the controlled plant consist of a model of building and working or operational limits of the process being controlled. Normally, the main goals of the model are minimum energy cost, dissatisfaction cost, and operational cost as well as tracking error.

The dynamic modeling required for the MPC controller must satisfy the following requirements (i) sufficiently accurate in describing system dynamics for a given set of design variables and (ii) computationally tractable allowing for real-time control and optimization. Trčka et al. [66] made an overview study on modeling methods for the HVAC control system and gave suggestions on modeling approaches relative to the simulation objective.

Kapetanakis et al. [67] investigated the effect of selecting input variables of data-driven models capable of predicting thermal loads of commercial buildings. The use of redundant input variables introduces unnecessary additional complexity during the development and execution of the models. ASHRAE [68] categorized modeling technique based on the estimation of energy usage: forward modeling (includes modeling of building and HVAC system and defining of optimization problem) and data-driven modeling (includes implementation of MPC and calculation of energy savings in the existing building and retrofitting).

A broader classification of modeling techniques is presented in Fig. 4. In general, three types of techniques are adopted for modeling building and HVAC systems that include physics-based modeling (Commonly referred to as white-box), data-driven modeling (Commonly referred to as black-box), and Hybrid modeling (Commonly referred to as grey-box model).

Description of physical characteristics (i.e. construction feature, building orientation, location) of building and HVAC system and its component are required to derive a physics-based equation. Heat and mass-transfer between thermal storage of buildings are designed with the help of the Resistance-capacitance (R-C) network. Commonly used laws for deriving mathematical equations of the above transfer processes include heat transfer law, mass transfer, momentum equation, heat

balance method, etc. The physics-based model has commonly applied techniques to model HVAC components. It one of the most suitable candidates for forecasting energy consumption and the performance of a controlled system. For example, Schirrer et al. [69] presented a novel idea of using a white-box non-linear model for heating and cooling system of a low-energy building. Patel et al. [70] captured the dynamics of a single cooling zone of the building through a physical model. Drgoa [71] implemented large scale white box-MPC in an office building for minimizing its thermal discomfort and energy usages. Zhao et al. [72] developed a baseline model with EnergyPlus software to maintain comfort criteria and minimize HVAC power consumption. In Ref. [73], R-C network was used for modeling to show thermal, light, and air quality dynamics for a single room. Fux et al. [74] compared several R-C networks to evaluate the model complexity required to capture the basic thermodynamics processes of house. Although these models have good colligation, they suffer from poor performance results. Thermodynamic processes are non-linear and cumbersome which leads to an increase in computational effort. The detailed physics-based model requires many iteration processes that may result in problems instability of the system and increased computational time. Such features are highly undesirable for online computation and BAS applications.

In the black-box model, input and output variables of the modeling system should be quantified as a mathematical relationship. Some of the mathematical techniques applied to the derivation of relationships are regression method, stochastic model, probabilistic model, a neural network such as ANN, and statistical method, etc. Experimental data are required for proper implementation while using this model. In comparison with the physics-based modeling, they require a fewer number of state parameters that help inaccurate prediction of future behavior. Chen et al. [75] used neural networks for predicting indoor operative temperature and air temperature based on external disturbances such as wind speed, relative humidity, season and office index, etc. Similarly, Chen [76] captured the dynamics of complicated physical processes i.e. indoor temperature, natural ventilation, and humidity inside the building using deep neural network. This transfer learning model

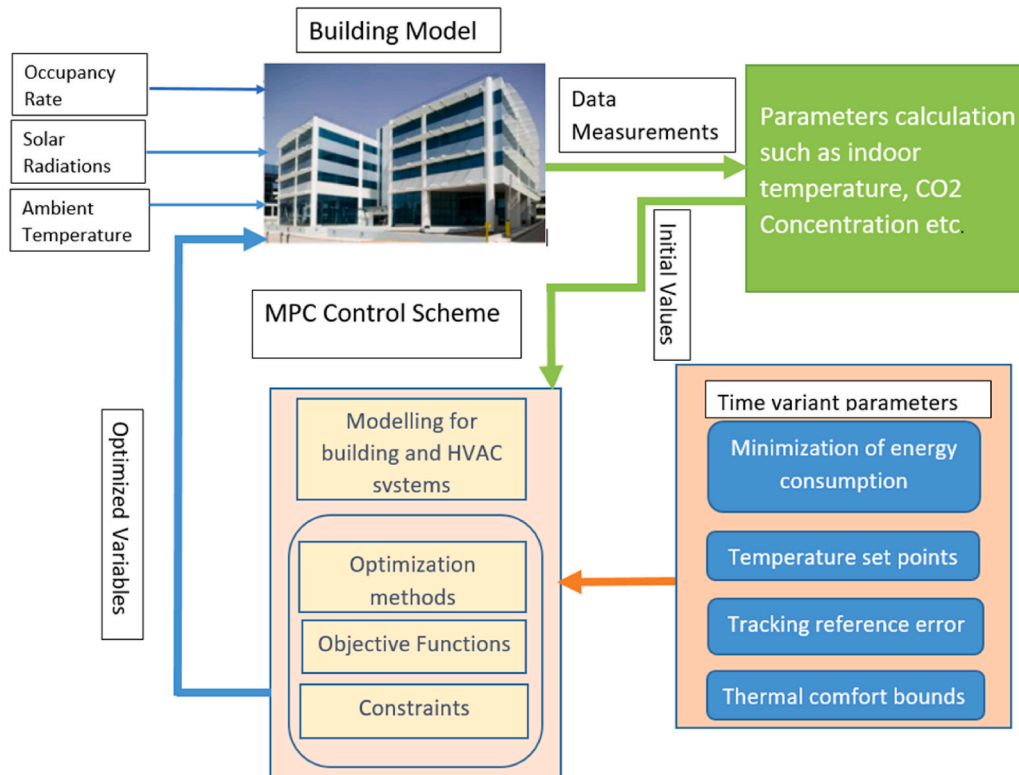


Fig. 3. Scheme of MPC control integration in building.

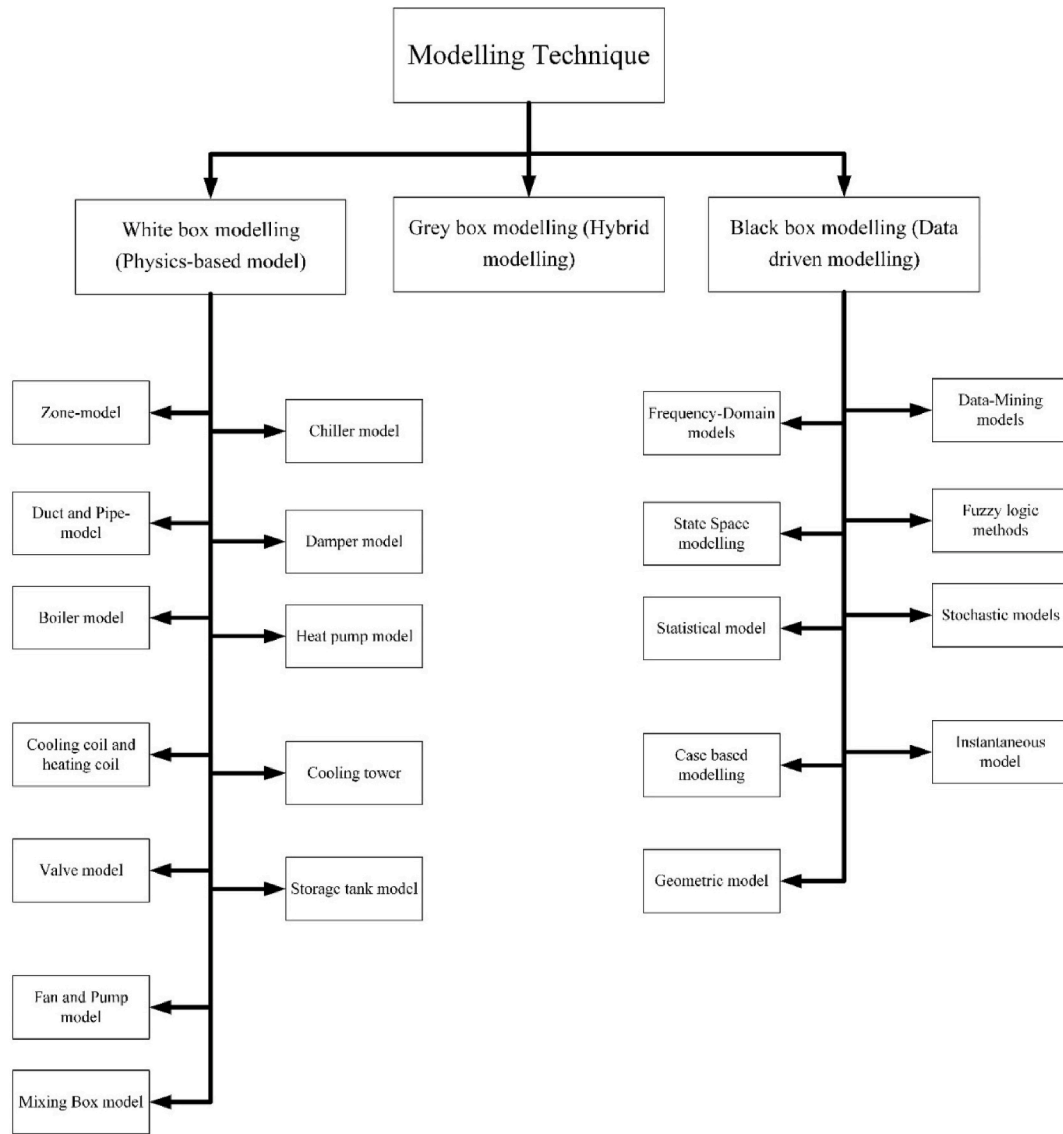


Fig. 4. Classification of modeling.

restricted the data requirement for modeling and design commissioning, thus reducing both cost and time.

Recently, there has been upsurge in works related to based reinforcement learning techniques. These techniques are broadly categorized into model-free reinforcement learning (MFRL) techniques such as Deep Q-network (DQN), Deep Deterministic Policy Gradient (DDPG), Trust Region Policy Optimization (TRPO), etc. and model-based reinforcement learning (MBRL). The model-free reinforcement learning (RL) requires extensive training with the environment for evaluating system dynamics. This extensive training leads to blockade in real-time implementation due to low efficiency. These techniques could not take into account of the constraints and problems involving constraints and cost functions are solved through penalty reward. These challenges can be overcome by implementing model-based reinforcement that trains its system dynamics with the help of neural network. Thus, MPC control scheme with the help of learned system dynamics is used to solve the defined optimization problem to obtain optimal sequence. Very few studies have been carried out on MBRL for precise control of building comfort (i.e. temperature, humidity control, CO₂ concentration etc.). Naren et al. [77] proposed a reinforcement learning-based (RL) controller for efficient energy management and climate control of buildings. The authors tried to implement the RL, MPC and RBC control

schemes for maintaining temperature and humidity constraints inside the buildings. The results showed that both RL and MPC controllers were effective in energy managements in comparison to RBC, and though the savings by RL is smaller than that by MPC.

Zhao et al. [78] tried to overcome the drawbacks of DRL as this mechanism is a model-free method that lacks learning efficiency and learning costs. This problem was addressed by introducing hybrid-model-based DRL for HVAC control problem. This hybrid-model-based approach utilizes both the knowledge-driven model and data driven model for whole learning process to reduce its learning cost. The simulated results showed that proposed method improves the learning cost during the process in comparison to model-free method. Ding et al. [79] implemented model-based reinforcement learning method (MBRL) to overcome the shortcomings of model-free reinforcement learning for HVAC control. The authors developed MB2C, a newly developed MBRL-based HVAC control system that can learn the building dynamics by employing an ensemble of environment-conditioned neural networks, and then learn MPC path integral for HVAC control with excellent sample efficiency. The results were executed in five-zone office buildings, and achieved 8.23% more energy savings compared to state-of-the art MBRL solution while maintaining same thermal comfort.

Li et al. [80] used subspace state-space system identification (4SID) methods for obtaining a simplified model for a predictive controller. Dobbs et al. [81] modeled occupancy behavior based on the two-state page Markov chain and later on was incorporated into HVAC control algorithms. Killian et al. [82] developed MPC scheme to exploit the full energy storage of smart homes and used a probabilistic model to predict the occupancy rate of smart homes.

These algorithms are particularly suitable for processing a large number of information in a very short period. Different types of black-box modeling techniques such as ANN [83–86], autoregressive with exogenous terms (ARX) [87,88], and NN [89–92] etc. have been commonly used in predictive control for building and HVAC system.

Grey-box modeling also known as hybrid modeling utilizes knowledge of the underlying physical phenomenon and input-output data of the system. The modeling techniques presented above have some of the advantages i.e. physical model provides good generalization whereas the black-box model provides better accuracy, such a feature can be incorporated in hybrid models. The grey-box model captures the thermal process dynamics of zoning inside the building through the electric analogy of the R–C equivalence network. The design parameters for this type of model need to be updated frequently while operating conditions deviate from training data for ensuring higher accuracy. State-space modeling is commonly used for this modeling technique as it belongs to a linear time-invariant (LTI) system. West et al. [93] designed a building model as a “grey-box model” for establishing the relations between ambient temperatures, achieved building zone temperatures and power usage of building HVAC systems. 3R2C modeling was used for linearly modeling the thermal dynamics of building for the DMPC controller in Ref. [94]. The air-thermal model was created for a hierarchical MPC control system having different periods of the horizon for energy storage systems and room level control [95]. Viot et al. [96] constructed grey-box modeling based on state-space resistance and electric analogy for anticipating control of long-time response floor heating system on the real building. Valenzuela et al. [97] modeled the dynamics of indoor temperature, incorporating closed-loop HVAC subsystems in the form of a discrete LTI system. Suda et al. [98] designed state-space modeling to capture the basic thermal behaviors between building and implemented MPC scheme to predict and control the future changes in the temperature of the room and the electrical charges.

The physical modeling technique does not have any requirement of a system prototype for energy analysis. Thus, it is the most proficient tool for the analysis stage and initial design. Even though the physical modeling technique is mostly used method for modeling, complexity arises due to their computing time. This may act as a hindrance for real-time application. Sound knowledge of physical phenomenon is required otherwise discrepancy in modeling might occur. The Black-box model has relatively high accuracy in comparison to the physics-based model but lacks of extrapolation of data beyond the training domain. This implies stable performance prediction cannot be ensured as an operational point extend beyond certain training data. This makes inverse modeling less feasible and adaptable in comparison to the white-box model. Grey modeling technique uses both features of white-box and black-box model i.e. require a sound knowledge of the physical process and data measurements. Thus, it seems to be a good agreement for modeling technique as it inherits the advantages of both white-box and black-box models.

Even though there are different methodologies for modeling techniques, the need for a perfect model still has not been met yet. Model validation in such cases seems to be an important step for better performance and results.

3.2. Building typologies and BMS management of HVAC systems

MPC control system has been applied for achieving goals like indoor thermal comfort, energy consumption in a building. These types of control systems have been implemented in a variety of buildings

irrespective of their architecture. The algorithm of the controlling system makes it feasible for energy management, indoor air quality, temperature setpoints, etc. in building despite their typologies. Experimental and simulation studies of this controlling algorithm can be achieved through field study or virtual building design.

Hanif et al. [99] implemented MPC control algorithm to calculate demand-side flexibility in large-sized office buildings. Different case study of this control algorithm is also based on the zoning of building i.e. single-zone and multi-zone in field of the HVAC system. Previous research studies on different typologies of buildings have been presented in tabular form and categorized in the form of residential [100], commercial, educational, etc. Educational building was selected for simulation and predictive control of temperature regulations and energy consumptions [101]. Reynolds et al. [102] designed a multi-zoning strategy in a small office building to calculate energy consumption and maintain thermal comfort. The single-story family house was modeled in TRNSYS for space heating and cooling process through predictive control.

The real-time application of MPC in the building seems to be more cumbersome in implementation. Building management system (BMS) using this control algorithm for the HVAC system requires to have three major software components for real-time data exchange and monitoring. The three major components are:

- 1) Interfacing Module: It is required to accommodate different BMS.
- 2) Data acquisition Module: It is responsible for communicating different data extracted from sensors, weather variables to the BMS interface module.
- 3) Control Module: It helps in executing model predictions, MPC algorithm and communication of control to BMS interface.

For BMS applications, they can be networked and controlled by center computers with the help of smartphones even walking through building complex. BMS improvise occupant comfort while reducing energy consumption by integrating with sensors such as motion sensors, CO₂ sensors, and HVAC timers to control lights, pumps, vents and heat exchanger in automatic mode or pre-defined values.

Framework for analysis of the potential BAS for performance optimization is presented [103]. This article presented the establishment of various methodologies that are not available in actual practice for building performance and simulations. Proper implementation of BEMS has led to energy-saving around 14.07% and 16.66% in the application of HVAC and other equipment [104]. This technological advancement in building management system has led to increased efforts on detecting fault diagnostics, and model-based control for building HVAC system.

One of the major concern is also due to the increase in failure of equipments and other misfortune leading to major loss in both from monetary and safety aspects. These are one of the serious concern that arises due to technological advancement in building management systems. This type of failures in automated control system arises due to comprehensible interconnected components, where localized faults occurs inside the sensors and actuators of the system. These type of errors in the system can led to severe damaging to the control process. Thus, fault-tolerant control (FTC) systems are studied with the serious concern that are capable of detecting and separating flaws in the BMS systems. In similar way, fault detection and diagnostics (FDD) have potential to reduce inefficiencies resulting from faults and degradation of building equipment and materials. FDD are basically categorized into two variants: Data-Driven based and Knowledge-driven based. Common FDD investigations are focused with vapor compression equipments like refrigerators, air conditioners, heat pumps, chiller and their inter-relationship with AHU. In general, common faults that occur inside AHUs are i) simultaneous heating and cooling (ii) heating valve leaking (iii) cooling valve leaking (iv) imbalanced airflows and (v) stuck dampers. Very few studies are conducted for FDD and FTC inconjunction with the MPC scheme. For ex, Adetola et al. [16] presented overall

development and application of MPC and FDD to large-scale HVAC systems, and implemented online. The authors implemented MPC algorithm using a new computational tool box the Berkeley Library for Optimization Modeling (BLOM), which generates automatically an efficient optimization formulation directly from a simulation model and the FDD algorithm uses heterogeneous sensor data to detect and classify in real-time potential faults of the HVAC actuators. The online implementations led to efficient diagnose of HVAC systems with energy efficient approach. Jarrou et al. [105] presented fault diagnosis and fault tolerant control based on model predictive control approach to achieve efficient zero energy buildings. The authors tried to achieve the optimum balance between energy production and consumption while reducing the total energy cost from power grid within defined constraints. The results showed that potential faults have severe effect on overall efficiency of the system and MPC seems to be most feasible control scheme for such abnormal operating conditions that makes it a good controlled for FTC. Subramaniam A et al. [106] implemented non-linear MPC for FTC and FDD for minimizing the monetary objectives while maximizing the overall occupancy comfort under the event of faults. This fault diagnosis system uses a full order non-linear to find and estimate multiple stuck in VAV dampers. The results showed that detection and estimation of faults lead with reconfiguration of MPC leads to significant improvement in energy efficient operation of HVAC in BAS. Shahnazari et al. [107] implemented recurrent neural networks (RNN) to develop models and a fault detection and isolation (FDI) methodology for HVAC systems. This approach does not require the historical data of plant system for building predictive models, and then embed within FDI filters. The simulated results showed that proposed FDI scheme is more effective than FDI scheme using subspace based models. This result has been verified through both simulation and experimental data cases. This approach has benefitted in both economical and safety concerns inside the building.

3.3. Constraints

MPC can explicitly deal with the system constraints, which enhances the embedding of physical problems involving constraints with the optimization methods. Constraints are categorized into hard and soft ones. The hard constraints should be satisfied without any compromise, i.e. it is mandatory. The soft constraints provide certain flexibility on operating limits and can be violated by incurring penalties in the objective function. During mathematical formulation, the hard constraints are generally expressed with equality sign, and the soft constraints are expressed with inequality signs. For example, Joe et al. [108] set all dynamics through equality constraints and bounds for the control inputs. The conditioned zone's temperatures were set as bounds on input and inequality constraint. Hard constraints were working limits of heat pumps, chillers, boilers, etc., and soft constraints were operational limits including temperature range, damper opening, CO₂ concentration, etc. Hou et al. [109] categorized constraints into local and shared ones. The local constraints were suggested according to thermal comfort requirements and HVAC requirements. The shared constraints involved different HVAC equipment and their corresponding agents. Different constraints were used for different goals. For example, in the study by Blum et al. [110], the room air temperature was controlled by imposing constrained value on zone air temperature and cooling thermal energy. Dobbs et al. [111] used heat flux flowing inside the zone as a constraint for controlling zone air temperature. Yang [112] restricted the normalized cooling power and PMV values to track the nominal set point temperature of room air reducing the energy consumption by 58.5%.

Some key points can be drawn from the previous articles as below:

- (1) Constraints play major role in achieving optimization goal and providing stability to feedback control system.
- (2) States of control systems are also directly affected through constraints

- (3) Constraints are basically expressed in the form of linear constraints (mostly used) that allow bounding limits on actuation, convex quadratic constraints that arise while bounding the total energy consumption among several actuators, and chance constraints mostly used with SMPC to limit in a probabilistic sense the behavior of system dynamics and input variables.

3.4. Optimizing functions and goals

For the MPC applications, the derivation of optimization functions is based on the objectives and behavior of the controlled system. These functions are generally defined and designed by researchers and energy designers in building HVAC systems. Previous studies have generally defined these functions using economic goals (e.g. power consumption and energy cost), thermal comfort goal (e.g. PPD and PMV), and control goal (tracking error and set-point). In particular, thermal comfort goals and economic goals are biased to each other. This can be better understood from the fact that users always expect the best indoor environmental comfort while hoping the minimum energy consumption. Thus, this problem can be addressed through defining objective function into a multi-objective function rather than a single objective function. This process is achieved by penalizing the cost function and adding some scalar terms to objective functions. Scalarization of multi-objective problems into scalar objective can be depicted as below:

$$Y(N_L, N_M, N_U) = \sum_{k=N_L}^{N_M} a(k)[g(t+k|t) - h(t+k)]^2 + \sum_{k=1}^{N_U} \Phi(k)[\Delta u(t+k-1)]^2 \quad (11)$$

where, g is the forecast of output; h is the reference signal; controlled effort is denoted by Δu ; N_L and N_M are the minimum and maximum prediction horizon and N_U is the control horizon; t is time step. Prediction horizon has the greater number than or equal to the control horizon. The terms $\alpha(k)$ and $\Phi(k)$ are sequences that will control the future response of the modeled plant.

These multi-objective functions mostly focused on energy consumption [33,109,113–123], thermal comfort Parameters (PPD, PMV etc.) [95–97], energy cost [124–127], both energy cost and demand cost [128–130], and deviation of indoor temperature [96,115,131]. Most of the objectives functions are particularly related to energy cost and thermal comfort parameters. For example, Biyik et al. [130] formulated MPC problem with economic objectives that included buying power from the grid, selling power to the grid, and cost operation of HVAC while achieving satisfactory thermal comfort. Álvarez et al. [132] implemented practical non-linear model predictive control (PNMPC) with the objective function of PMV for regulating the building's temperature. Feng et al. [123] implemented MPC scheme to optimize the water valve position in order to minimize the weighted combination of comfort violation and energy usage over a defined prediction horizon. Bahramnia [133] optimizes minimizes the energy consumption and maximizes thermal comfort by tracking the error of two interactional channels i.e. temperature and humidity.

Some of the major points that can be drawn from past literatures are presented below:

- Minimization of energy and power consumed by the HVAC system can be expressed as objective function neglecting the thermal comfort.
- Expressing objective function in terms of tracking error can be useful if the tariff rate of energy consumption remains constant and if there's no onus on energy saving.
- Non-linear objective functions increase the computational time for the optimization algorithm.

It has been generally seen that the objective function of an optimization algorithm plays a vital role in finding an appropriate solution to the given problem along with properly defined constraints. Proper formulation of the problem ensures less computational effort.

3.5. Horizons and sampling time

Prediction horizon of MPC refers to several future control intervals required for forecasting while optimizing the manipulated variables at the control interval by the MPC controller, which is an important term that greatly attributes the performance of the controller. The control horizon refers to several manipulated variables to be optimized at the control input. Control horizons are always limited between 1 and the prediction horizon, i.e. they have smaller value in comparison to the prediction horizon. The values of the prediction horizon and control horizon were assumed to be the same in many studies. For example, Zakula et al. [134] proposed MPC scheme with a target of minimizing total cooling energy consumption, and with the planning and control horizon of 24 h. The results validated their values directly affected computational times that are extremely relevant for real-time implementation and are dependent on the accuracy of weather and load predictions. Fang et al. [135] derived MPC-based ventilation rate curves with 12-hrs prediction horizon and 12-hrs horizon over the defined period. Mei et al. [136] simulated MIMO MPC for single-zone building with prediction horizon and control horizon set to 24 and 24 h, respectively for control objective of minimization of energy consumption. The longer prediction horizon results in an increase in sampling time and ultimately leads to an increase in weight of cost functions whereas the smaller control horizon needs fewer computation resources and makes the controller more stable. Sampling time co-relates with a time constant of the system which means the control variable remains unaltered over that period. The lower sampling time improves the rejection of unknown disturbances but adversely increases the computational effort. Dynamic order of controlled structure must be chosen with 10–50 times the sampling time, which may give rise to a great number of parameters. HVAC systems, in general, have a higher response rate which eliminates longer sampling time and handles disturbances factors in more efficient ways. For example, Yang et al. [137] investigated three sampling times of 1 min, 2 min, and 5 min during the implementation and tuning stage of MPC. The result suggested sampling times (control interval) of 2-min was capable enough to deal with dynamic occupancy level whereas the longer control interval i.e. 5-min could not accommodate the dynamics of indoor occupancy load. On the other hand, it was observed that the control interval of 1-min was unable to track the cooling set-points generated by MPC due to change in frequent values. Sangi et al. [138] assumed a prediction horizon of 10 min with a sampling time of 2-min for real-life application as hot water storage could be fully loaded by the heat pump during this period. The results suggested that the average calculation time of optimization was greatly reduced, and therefore faster data communication can be established. Ul Haq et al. [139] simulated a dual-mode model predictive linear controller (DMMPLC) scheme having a horizon of 4s and the total simulation time of the 90s was considered for studies on trajectory tracking of discrete non-linear dynamic systems. The results obtained from simulations suggested that setpoints were quickly achieved with minimal energy consumption. Gao et al. [140] adopted a sampling interval of 15s to control the space temperature of the middle-sized room cooled by Variable air volume (VAV) type of air-conditioning system. The results showed that a smaller value of sampling time leads to the fast response of the controller. Benga et al. [141] presented the idea of updating sensor measurements and weather forecast, estimating temperature states, diagnosing component faults, generating optimized set-points for the entire prediction horizon for 4 h. Rawlings et al. [142] performed an optimization problem simulating it for one week of predictions having a time step of 1 h for low-level waterside subproblem, and 15 min for both high-level and each of the low-level airside

subproblems. The results showed that computational time for smaller time step (i.e. 15-min) was lesser than greater time-step (i.e. 1 h) and decomposition of sub-problems could also be implemented online since control executions occur every 15 min. The exhaustive simulation was conducted with varying prediction horizons between 1 and 30, the sampling time of 3 s, and the control horizon having a value of 10 to achieve a set of final tuning parameters that give desired smooth close-loop performance in Ref. [49]. The results suggested smaller values of the control horizon were required to obtain the undesirable transients in the distributed algorithm.

Following points can be summarized from the literature:

- (1) Length of prediction horizon should be assumed in such a way that it is long enough to capture the transient dynamics.
- (2) Thermodynamic processes in buildings are found to be steady and have the longer settling time
- (3) Horizons and sampling time are usually defined by the behavior of the processes being controlled and time-step length.
- (4) Time step length and sampling time during optimization process should be reasonably short so that MPC can adapt to these changes. But, it should be kept in mind that the shorter control interval will result in more interval over a prediction horizon, and ultimately lead to burden in solving optimization process.
- (5) Short horizons are preferred because of their flexibility in the computational process, and the number of variables for the optimization problem algorithm gets reduced.
- (6) Closed-loop (CL) feedback-controlled processes have a desirable feature of improved performance and stability. Thus, long horizons are preferred for closed-loop operation. Sometimes operational time is infinitely far in the future. This may be termed as an infinite horizon case which may lead to constant feed-back gain matrix.

4. MPC implementation

Implementation of MPC in HVAC systems needs a lot of work. To begin with, energy consumption and occupancy behavior of buildings are required to be recognized. Then, the optimization method and disturbances factors are extensively assessed according to real situations. Before the MPC controllers are carried out in a real system, they are often investigated with the help of some softwares.

4.1. Optimization methods

Optimization algorithms play a vital role in the real-time implementation of MPC in a short period.

Optimization methods are to find the best solution among many possible alternatives without exhaustive search and comparison. They are highly dependable on the function of control variables (e.g. temperature, humidity, and pressure etc.) and objective functions related to economic goals and thermal comfort for MPC control schemes. Basically, these methods are categorized into two types, i.e. linear and non-linear optimization ones. The linear optimization technique is a class of optimization where the objective function to be optimized and all the constraints are linear in terms of the decision variables. This technique is further sub-classified into a direct approach, iterative approach, recursive approach, etc. The non-linear methods are complicated and more complex in solving and finding out the solutions in comparison to the linear methods. This type of method is sub-classified into direct search, gradient-based and non-linear global optimization techniques that include genetic algorithm, simulated annealing, and evolution strategy, etc. A broader classification of these optimization methods is illustrated in Fig. 5.

Non-linear thermodynamic processes are generally modeled into linear time-invariant (LTI system), and objective functions of such systems are mostly solved using LP and QP algorithm based on the degree of

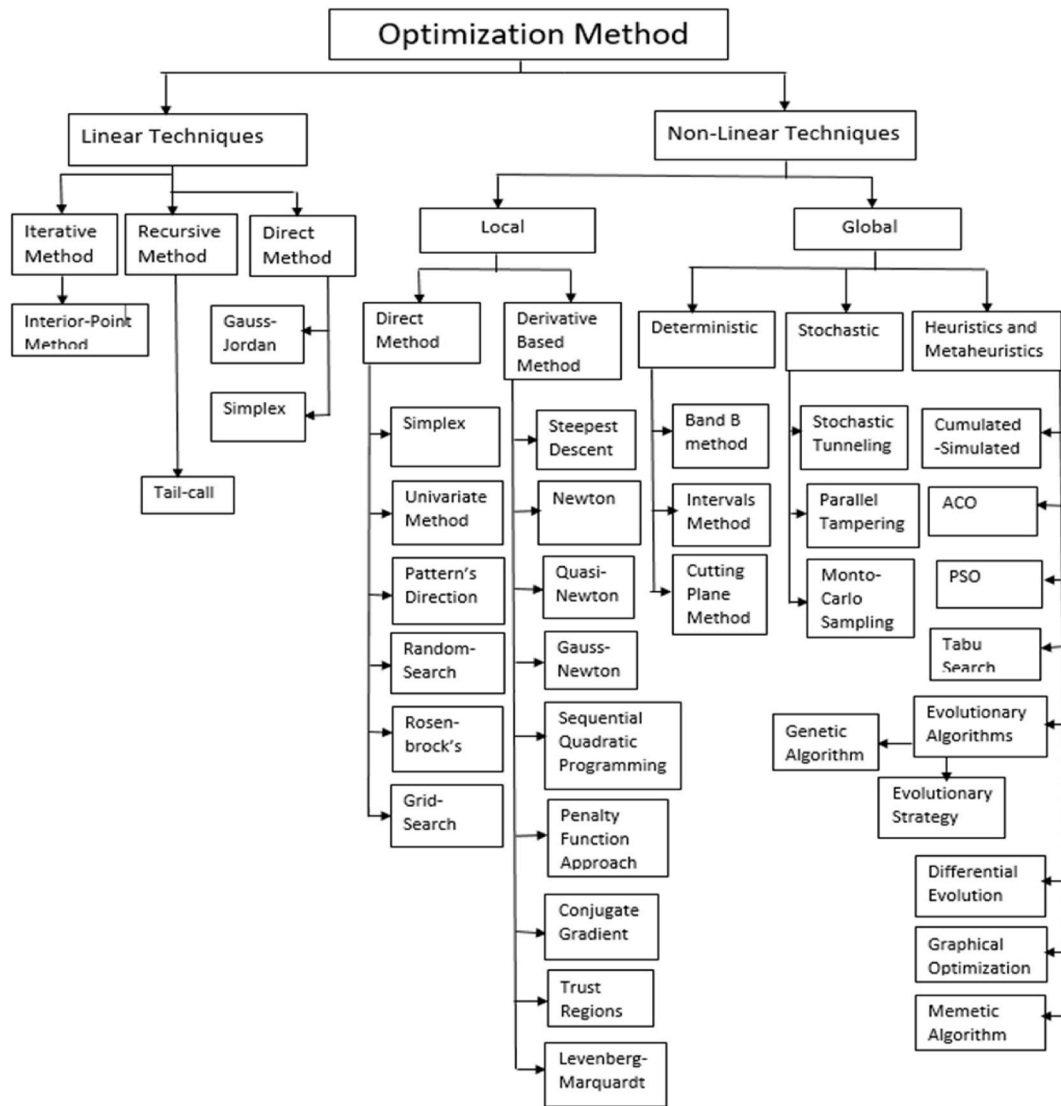


Fig. 5. Classification of optimization methods for MPC application.

variables (i.e. highest power of variables) of these functions. Vasak et al. [143] used linear programming techniques to find a sub-optimal solution that keeps zone temperature is given boundaries and appropriate cooling/heating power. Similarly, Bălan et al. [144] first formulated a mix-integer nonlinear constraint optimization of system energy and then employed the same optimization algorithm i.e. LP to solve the optimization problem. Huchuk et al. [145] implemented quadratic programming to assess the energy-saving potential of an office over prediction horizon of 24 h. Luzi et al. [146] defined optimization problem to track optimum temperature set points while minimizing energy consumption inside the building. The authors expressed the problem in standard QP form and solved it using QP solvers. Sturzenegger et al. [147] proposed sequential linear programming (SLP) to minimize operating costs ensuring thermal comfort for the bilinear system. All of the objective functions defined in these articles were defined with single optimization goal. During the last decades, there has been a use of multi-objective optimization functions rather than single ones.

Rise of multi-objective optimization and their trends has been reviewed graphically in Ref. [148]. Several heuristic optimization methods such as bio-inspired algorithms, evolutionary algorithms, etc. were summarized in this article. Dullinger et al. [149] defined multi-objective functions involving the cost functions such as power consumption, penalizing deviations of CO₂, enthalpy, net mass flow, and

switching of HVAC configuration and utilized Mixed-integer linear programming (MILP) algorithm to optimize these functions. This article suggested that the applied optimization technique approximated global optimality and decreased the computational techniques. Separately, Gayeski et al. [150] designed a general pattern search algorithm (GPS) to efficiently solve the chiller control sequence every hour and reduce energy consumption of cooling system over 24-h look ahead moving horizon.

Another two commonly-used optimization algorithms are particle swarm optimization algorithm (PSO) and genetic algorithm (GA). The PSO algorithm is essentially meta-heuristic search technique, which is generally being adapted and applied to the HVAC field to find near-optimal solutions. For example, Corbin et al. [151] employed PSO in receding horizon MPC scheme to minimize electricity building in residential HVAC systems. This algorithm has simplified rules, independent of derivatives, optimizing functions, and can operate on black-box modeling having discontinued spaces. The GA technique has been used as an optimization tool for minimizing the energy and demand having non-linear constraints. Hilliard et al. [152] determined zonal air temperature setpoints using genetic algorithm for whole building energy minimization optimization. Ascione et al. [153] implemented genetic algorithm in simulation-based MPC to optimize the hourly set point temperatures with the day-ahead planning horizon to calculate the

multi-objective optimization of operating cost for space conditioning and thermal comfort. Since the GA method is a population-based optimization technique which makes it furthermore suitable for solving multi-objective optimization(MOO) problems.

4.2. Disturbances in MPC

The discrepancies that leads to obstacle in real-time implementation of MPC control in building and HVAC system arises are mainly due to model mismatch. This model uncertainty in MPC control scheme arises due to uncertainty in parameter factors and external disturbances. In general, parameters uncertainties in plant modeling arises due to appropriate equations depicting the dynamics of the system. The error measurements in external disturbances such as ambient temperature, solar irradiation, wind speed, occupancy rate, inaccuracies in measuring device, and other unmeasured disturbances.

Disturbance modeling in the MPC scheme helps to achieve offset free-control and stable system. This scheme has one exciting feature of satisfying state constraints even when set bounded disturbances are considered. Thus, this scheme calculates future states system based on the estimation of disturbances acting on the systems. Furthermore, there is a requirement to fill information on feedback values of states and disturbances to linearize model and disturbance decoupling. Therefore, sound knowledge on disturbances is inevitably required to reduce uncertainty in the designed model that poses serious challenges for real-time implementation of MPC in building HVAC systems. Basically, disturbances are classified according to their source of origin, namely external and internal disturbances. In general, external disturbance includes climatic factors such as wind speed, atmospheric temperature, humidity, radiation, insolation, etc. Internal disturbances include utility cost cooling load, thermal bounds, temperature setpoints, damper positions, occupancy, etc. For example, Sayadi et al. [154] used eleven parameters of disturbances for model designing. These disturbances were the temperature of adjacent rooms, ambient temperature, solar radiation, internal heat gains, supply heat from concrete core conditioning to system floor, mass-flow rate of the high-temperature air stream, surface area of windows and walls, emissivity of walls, heat capacity of air, Stefan Boltzmann constant for radiation and shape factor for irradiate heat. Separately, Schwingshackl et al. [155] considered disturbances such as hot water temperature, cooling temperature, and excitation signals for controlling temperature and relative humidity. Dullinger et al. [149] optimized and maintained thermal comfort considering environmental disturbance factors such as solar radiation and ambient temperature. In another study, Gruber et al. [156] utilized both external and internal disturbances which included people, lighting equipment, door opening, solar heat gain, and outside air temperature to study the MPC control algorithm. A major challenge of implementing MPC lies in the prediction of disturbances due to their stochastic nature. Maasoumy et al. [157] proposed two methodologies for handling model uncertainty for MPC. The authors proposed a modeling framework for the online estimation of states and unknown parameters leading to the parameter-adaptive building model. This article further proposed robust model predictive control (RMPC) formulation to make a building controller robust to model uncertainties. The RMPC was also compared with nominal MPC and RBC. The results showed that the RMPC was found to be superior to an intermediate level of uncertainty (30–67%), while nominal MPC was preferred for the cases with a low level of model uncertainty (0–30%). It has also been seen that RBC outperformed MPC or RMPC when model uncertainty goes beyond the threshold (67%). Several literature discussed above gives us the idea of incorporating different disturbances in MPC implementation. Still, methods of data gathering about disturbances remain unanswered. Particularly, they are gathered through offline methods that include onsite measurements, cooling load charts, modeling through observations etc., and online methods that include weather information from the meteorological site. The online prediction method is more reliable and provides an accurate

result for complicated models although it requires internet communication as opposed to offline methods. For instance, Beghi et al. [127] applied the ANN algorithm to obtain accurate prediction by exploiting a limited data set of standard measures available in HVAC installations. Input predictions were used as input of the ANN model to forecast 24-h cooling load profile. In addition, Darivianakis et al. [158] utilized an offline prediction approach to calculate weather forecast from COSMO-7 of MeteoSwiss and obtained forecasts of internal gains from standard weekly profiles of typical building configurations.

In summary, techniques of compensating for unmeasured disturbance are modeling, measurement, and inputs that minimize the effect of disturbance on controlled variables. Their reliable selection is necessary to achieve zero-offset goals.

4.3. Current softwares for MPC study

Building energy simulation software is the potential tools for energy and economic analysis of HVAC in buildings. As depicted in Fig. 6, this software normally provides a simulation platform for building modeling and HVAC system modeling based on which some advanced control algorithms like MPC can be studied.

These softwares primarily aim at proper sizing of the HVAC system, prediction of energy consumption pattern, monetary objective, thermal behavior of building, etc. In particular, desirable features of these simulation softwares are open source, friendly graphical user interface (GUI), co-simulation accessibility, intact training manual, HVAC modeling components, and so on. Atam et al. [159] reviewed the topic on these software barriers to advanced model-based control design for energy-efficient buildings. This paper illustrated commonly used softwares and tools and presented advanced model-based control design for energy-efficient buildings. Meanwhile, a novel concept of functional mock-up interface and associated advances were reviewed in detail.

Various simulation software can be readily available. Most used modeling software packages include TRNSYS [160–163], MATLAB [164–167], EnergyPlus [168–172], ESP-r [173], DOE-2 [174], IESVE [175,176], Modelica [177–181] etc. There have been several studies that reported significant improvements in energy efficiency using these softwares for MPC schemes. Zhang et al. [182] used TRNSYS simulation to model two residential buildings each having five zones to reduce the electrical demand for heating software through the MPC control scheme. This article further coupled building model with Genopt to compare different optimization algorithms. In addition, Sourbron et al. [183] modeled a two-zone office building with concrete core activation (CCA) and the HVAC system for maintaining the indoor environment and reduce energy cost with TRNSYS. Furthermore, Wallace et al. [184] carried out whole year simulation of an office building, and results were validated with the help of EnergyPlus. Aftab et al. [185] designed large public mosque for indoor climate control resulting in energy savings using EnergyPlus, and real-time data exchange was done with the help of BCVTB. Blum [178] developed single-zone office building using Modelica and used the same software for compiling models into the functional mock-up unit (FMU), simulating the FMUs, and solving the dynamic optimization problems for parameter estimation and optimal control. Drgoña et al. [180] created house with six zones and modeled with high accuracy using the open-source Modelica and later transformed into LTI model. These softwares can be used with third-party user interfaces, such as GenOpt [186], BCVTB [187,188] and etc., for solving optimization problems and real-time data exchange. The objective function of these optimization problems cannot be directly evaluated with modeling softwares. Optimization solvers, such as CPLEX [189–191] and YALMIP [192–194], are available in either open-source or commercial form for obtaining the global extremum value of these objective functions. One of the main concerns of using these types of softwares is that it cannot always guarantee accurate results.

In summary, simulation models are free of space and time

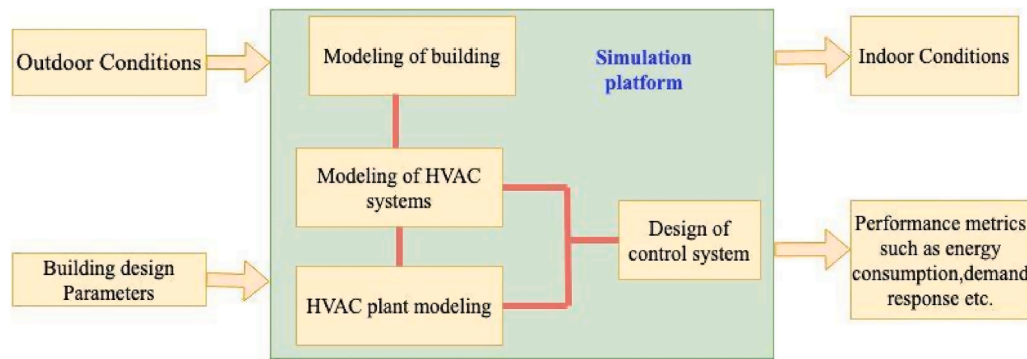


Fig. 6. Schematic diagram for energy simulation.

limitations. However, their applications are limited to high-level supervisory control. Building characteristics and constructions are subject to continuous variation according to client demand. High computational process and software expertise is required for such softwares resulting in high cost.

5. Summary

5.1. Conclusions

Critical reviews of previous articles has focused on providing a complete problem formulation of optimal control scheme for efficient energy operations inside the buildings and their HVAC systems. In this paper, comprehensive analysis has been made in terms of its implementation, optimization algorithm, its application, and modeling as well as overall picture related to the MPC control schemes. The findings of the review paper can be summarized in as follows:.

- (a). The modeling technique is considered one of the major crucial steps while implementing the MPC control scheme for building and HVAC control. From previous studies, the state-space model (LTI) has been mostly used for modeling. There have been no straight forward (silver bullet) solutions for a perfect model match. Different types of black-box modeling such as ARMA, ARMAX, ANN, etc. are being used in increasing trends. Complex building models that contain thorough information on building behaviors and dynamics are designed through white-box modeling techniques and simulation tools such as DYMOLA, TRNSYS, Modelica, EnergyPlus, etc. Even though, black-box models are in an increasing trend for application of modeling, physical and hybrid models are seen to have a certain edge in comparison to them. Physical and hybrid models are more beneficial while designing a model as it requires less information and expert knowledge. Validation of the model is also required when there is a certain degree of model mismatch. Very few previous articles have considered model validation and the degree of errors associated with them in their research works.
- (b). One of the main issues while implementing a control scheme for building applications is to balance conflicting aims i.e. maximizing thermal comfort while minimizing the energy consumption and costs. Building dynamics are non-linear, so the global optimized value is difficult to obtain and requires perfect mathematical formulations of a problem that are complex. Although a lot of previous studies have generally used linear and quadratic programming techniques for solving optimization goal, various non-linear techniques such as GA, PSO, Hookes-Jeeves method, MILP, Branch and bound method, etc. are being extensively used for resolving multi-objective goals. Real-time application of complex optimization techniques may result in longer computation time.
- (c). Proper implementation of MPC requires well-defined numerical values of control horizon, prediction horizon and sampling time. These factors directly affect computational efforts, the stability of controller, rejection capability of disturbances factors, and setting time of MPC controller. Previous studies have shown that sampling time is considered in the span of 5min -1hr and prediction horizon of 24 h in general. Some of the research scholars have also considered large sampling time like 6 h and prediction horizon greater than 24 h such as 48 h, 96 h, etc. Weather predictions and other climatic disturbances are generally updated at 1 h Interval, thus in such cases above assumption of sampling time and prediction horizon can be justified.
- (d). Energy management and savings strategy are the main objectives for the modern architectural model of buildings. Nowadays the building is designed with expendable integrands when solar pumps and collectors are being used. BMS applications generally use a hybrid model i.e. combining two or more HVAC components at a time such as a solar cooling system with heat pumps, chillers being used with ventilation systems, etc. Previous studies have considered the study of building with thermal storage such as PCM material, photovoltaic material, storage tanks with various HVAC components such as chiller, boilers, heat pumps, etc. These types of studies are generally formulated under mixed-integer problems and solved with help of optimization techniques like MILP and MIQP etc. MPC in conjunction with TES enables in load shifting to off-peak hours to reduce peak demands that help in further reduction of energy consumption and savings MPC provides better results in energy savings when compared to heuristically rule-based controllers and other control schemes.
- (e). Smart building with full automation is a future candidate for the next type of building design. Thus, MPC stands out to be one of the promising candidates for the control scheme as it balances out conflicting optimizing goals and downsizing of cooling and heating loads. Although there has not been extensive research in a residential building with automation there is certainly a sharp rise of study in such types of buildings. MPC outperforms other controllers for investigating, research and future applications in smart buildings as it is more robust and handle disturbances factors in efficient manners.
- (f). The performance of the MPC controller can be enhanced by better estimation and prediction of disturbances factors such as humidity ratio, ambient temperature, solar irradiation, occupancy rate, etc. Previous data can be collected for more proper information about the behavior of these factors through BMS application.
- (g). Previous studies have generally conducted their simulation or experimental studies of building HVAC systems such as residential, commercial educational, etc. BMS can be easily incorporated in such types of buildings with MPC as it has better technicality. Some of the major issues hindering the MPC market's growth in

real-life systems are the lack of skilled and efficient experts, the cost of retrofitting, etc.

- (h). MPC outperforms other conventional controllers such as PID, RBC, PI, heuristic control, etc. In terms of energy-saving, minimizing cost consumptions, maximizing thermal comfort, etc.

5.2. Future outlooks

MPC is still a wide-open subject, and much work still has to be done in this field. Several areas that are yet to be explored for reliable and better performance of MPC are listed as below:

- Perfect software for the MPC control scheme in one platform that includes modeling, optimization, monitoring, and prediction of disturbance factors is still lacking.
- Even though there has been wide use of different kinds of non-linear optimization techniques, still different types of techniques are yet to be explored. Some of the optimization algorithms that can be utilized for future work and better performance of MPC are BAS, ACO, various kinds of GA algorithms, bacterial foraging, Harmony search, clonal selection, differential evolution, immune algorithm, imperialist competitive algorithms, cat swarm optimization, etc.
- Future work can be focused on better prediction of disturbance factors such as occupant behaviors, weather forecasting for improvement in MPC performance.
- MPC integrated with AI techniques can be future research goals for smart buildings.
- Different kinds of MPC such as AMPC, DMPC, SMPC, CMPC, RMPC, H MPC, etc. have been considered in previous studies separately, but there is still a lack of comparative study between them
- Design parameters such as prediction horizons, control horizons, and sampling time which directly affect the MPC control scheme need to be studied in detail.

Declaration of competing interest

There is no any conflict of interest.

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